Mapping the borderlands

Comparing two encyclopedias in the Polish-Ukrainian borderland

Context

For this project I want to use AI to research a field that I have a long-standing interest in and that I have much personal experience with. I want to research the demographic situation in the former crown land of Galicia (modern day south-eastern Poland and western Ukraine) in the 19th and early 20th century. During that time this region was a multicultural melting pot of many different cultures (Poles, Ukrainians, Jews, ...). The fall of the empires and the establishment of nation states was looming at that time, so the question of where national borders should be established was highly controversial.

Sources

I want to use the *Gemeindelexikon* (Austrian census data from 1900) and the *Słownik geograficzny królestwa polskiego* (Polish geographic encyclopedia) and compare statistical data of the two sources with each other using LLMs, both are available online.

Questions

Q1: How does the *Słownik* describe east Galician villages as compared to census data?
H1: *Słownik* entries have a bias toward polish and catholic elements.
H2: Where language or religion diverge, *Słownik* resolves this ambiguity by representing given communities as Polish.
Q2: Do LLMs give different results when given the same prompt in different languages?
H3: The LLM will have a bias towards the nationality of the language it was given the prompt in.

Why it matters and why use AI?

The *Słownik* and the *Gemeindelexikon* are still major sources for historians studying the area, although they are known to diverge in some areas. The *Gemeindelexikon* has a lot of statistical data, that means numbers and graphs, the *Słownik* however gives the reader textual descriptions of towns and the people living in them, so they are hard to compare side by side. Using LLMs however it is much easier to summarize and quantify the *Słownik's* long texts and compare them with statistical data. Therefore, I want to build a RAG system over both of them, so that every answer shows which entry it comes from.

The AI's ability to understand different languages also comes in handy, since we are dealing with sources written in different languages, luckily, I also understand and speak the languages relevant to this area, so I can double-check the AI's work. I also want to ask the AI for summaries of the data in different languages, seeing if the results vary.

Concerns

There are unfortunately no comparable contemporary Ukrainian or Yiddish sources present on this topic, so these voices are not fully represented. I will still try to give Ukrainian prompts for Q2, to see if results vary. Also, reading the texts might be difficult for AI, as there can be errors in the OCR data, therefore proofreading might be necessary.